

Housing Data Analysis & Predictive Modeling Report

A clear, business-friendly summary of the most important findings from the housing dataset, including exploratory analysis, model evaluation, and practical next steps.

ROWS

900

COLUMNS

82

AVG. SALE PRICE

\$192,204BEST R²**0.917**

What I Focused On

The work is organized around two main parts of the assignment: exploratory data analysis and predictive modeling for **SalePrice**. The goal was to understand what drives house prices first, then build and explain a model that non-technical stakeholders can follow.

Analysis Approach

I treated this as a real housing analytics problem: check the data quality, identify pricing patterns, then move into modeling. This makes the final model easier to trust because the results are connected back to business meaning.

Workflow Summary

1

Load data

2

Check quality & missing values

3

Explore price drivers

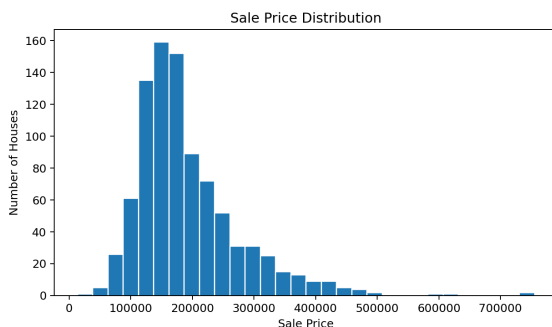
4

Train regression models

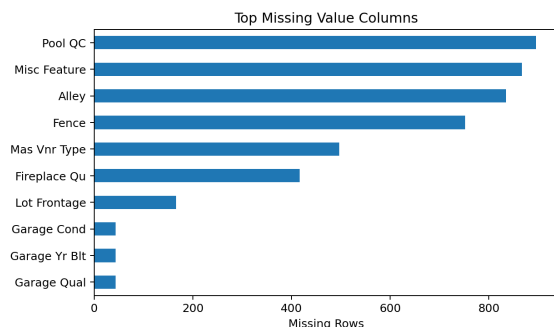
5

Evaluate & explain results

Sale price is right-skewed



Missing values show property feature gaps



Dataset Quality Notes

The dataset contains **900** rows and **82** columns. There are **25** columns with at least one missing value. Some missing values likely mean the property does not have that feature, such as a pool, alley, fence, or fireplace.

Modeling decision: Numeric missing values were filled with the median. Categorical missing values were filled with the most common category as a clean baseline.

Top Missing Columns

Column	Missing Rows
Pool QC	896
Misc Feature	867
Alley	835
Fence	752
Mas Vnr Type	497
Fireplace Qu	417
Lot Frontage	166
Garage Qual	44

Key Business Findings

Quality matters most

Overall quality had the strongest numeric relationship with sale price. Higher-quality homes show a clear price premium.

Size is a major driver

Living area, basement area, first-floor area, and garage area are strongly connected with higher prices.

Location changes value

Neighborhood differences are large, so location should be treated as a key business feature.

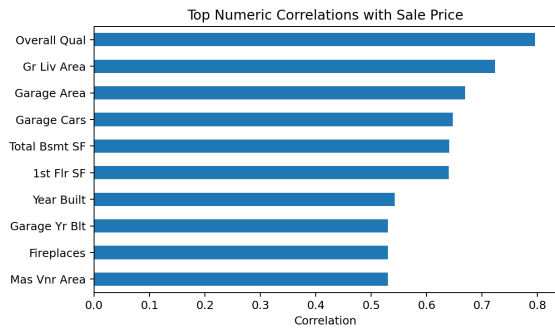
Overall quality creates a visible price premium



Living area is positively related to sale price



Strongest numeric correlations with SalePrice

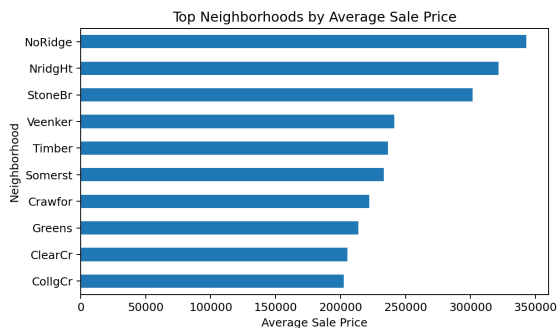


Top Price Drivers

Feature	Correlation
Overall Qual	0.795399
Gr Liv Area	0.723209
Garage Area	0.669825
Garage Cars	0.647759
Total Bsmt SF	0.640525
1st Flr SF	0.640095
Year Built	0.542092
Garage Yr Blt	0.530598

The strongest signals are practical real-estate factors: quality, space, garage capacity, basement size, and newer construction.

Neighborhood price differences



Top Neighborhoods

Neighborhood	Rows	Average Sale Price
NoRidge	31	342985
NridgHt	74	321494
StoneBr	21	301715
Veenker	16	241481
Timber	18	236580
Somerst	92	233175

Neighborhood price ranking should be read carefully when row counts are small, because one or two houses can distort the average.

Predictive Modeling

The modeling task is a regression problem because the target, **SalePrice**, is a continuous numeric value. ID-style columns were removed, prepared numeric and categorical features separately, and compared multiple models using the same train/test split.

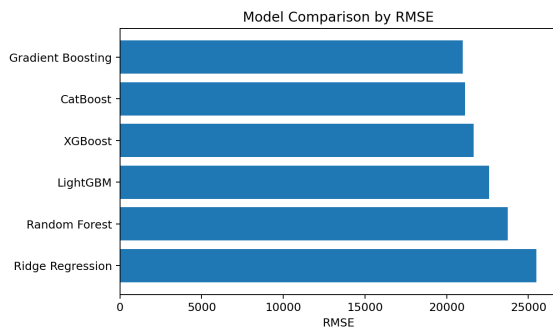
Train/Test split: 80/20

Target: SalePrice

Metrics: MAE, RMSE, R²

Best model: Gradient Boosting

Lower RMSE means better price prediction



Model Results

Model	MAE	RMSE	R2
Gradient Boosting	14139.05	20995.50	0.9174
CatBoost	14722.30	21128.86	0.9164
XGBoost	14469.82	21664.71	0.9121
LightGBM	14384.82	22606.94	0.9043
Random Forest	15304.19	23750.61	0.8943
Ridge Regression	17746.27	25490.04	0.8783

Final Model Choice

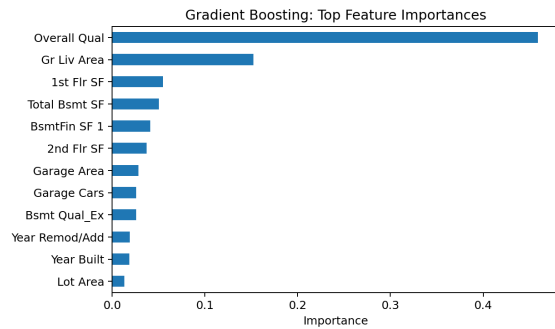
Gradient Boosting was selected because it gave the strongest result on the test set with an RMSE of **20,995.50**, MAE of **14,139.05**, and R² score of **0.9174**.

In plain terms, the model explains about **91.7%** of the variation in house prices on the test data.

Why This Model Makes Sense

Boosting models work well for structured housing data because they capture non-linear relationships. For example, the value of a bigger garage or larger living area may depend on the neighborhood, quality, and overall condition of the house.

Important model features



Evaluation Meaning

MAE shows the average dollar error. **RMSE** penalizes larger mistakes more, which is important for expensive properties. **R²** shows how much price variation the model explains.

This gives both a technical and business-friendly view of performance.

Potential Improvements

Better missing-value meaning

Replace some missing values with labels like No Pool, No Garage, No Fence, or No Fireplace.

Feature engineering

Create practical features such as house age, years since remodel, total bathrooms, and total finished area.

Model tuning

Use cross-validation and hyperparameter tuning for Gradient Boosting, XGBoost, LightGBM, and CatBoost.

Optional Recommender System Idea

The same housing data can support a basic real-estate recommendation engine. If a user likes a house, the system can recommend similar homes using price range, neighborhood, living area, bedrooms, bathrooms, garage size, and quality score.

Similarity search

Budget filters

Neighborhood preference

Personalized listings

Expected impact: users find relevant homes faster, engagement improves, and the platform can increase saved listings and agent contact requests.

Final Conclusion

The analysis shows that housing prices are strongly influenced by quality, living area, garage capacity, basement size, house age, and neighborhood. The final model provides a strong first version for predicting sale price, while the business insights explain why those predictions make sense.

The next version should focus on richer feature engineering, clearer missing-value treatment, and cross-validation to make the solution more reliable before production use.